TAYLOR INVENTION

Thermally Controlled Plant Nursery Assembly — Investor Pitch Brief

"Securing the global high-value crop supply chain against climate change through micro-climate root insulation."

1. Executive Summary

Warming global temperatures are shrinking the geographic sweet-spots required to grow high-value, temperature-sensitive crops like specialty Arabica coffee and premium botanicals. Traditional nurseries are vulnerable to pathogen contamination, external weather shocks, and high energy costs.

The Taylor Invention (Patent No. US 2024/0407307 A1) solves these vulnerabilities by isolating individual plants within a self-contained, automated, mobile micro-climate system. By cooling the soil and root zones directly through an integrated cooling-coil network, we achieve unmatched thermal efficiency and biosecurity, enabling cultivation anywhere on Earth.

US 2024/0407307 A1
Granted/Published Patent

100% Isolated
Biosecure Air & Water

Off-Grid Ready
Solar & Generator Inputs

2. Key Investor Value Propositions (The Moat)

- **Direct-to-Root Thermal Cooling:** Unlike standard greenhouses that waste vast energy cooling massive rooms, this system embeds cooling loops directly into the container sleeves to target root-zone temperature—the primary driver of plant health and metabolic rate.
- **Integrated Pathogen Protection:** A built-in array of ultraviolet, blue spectrum, and fluorescent sanitizers sterilizes air, water, and interior surfaces to prevent early-stage crop loss due to mold, fungi, or bacteria without utilizing chemicals.
- **Patented "Breathing" Containers:** A nested, dual-cylinder design lets operators manually rotate nested buckets to align precise ventilation slots, adjusting root aeration dynamically.
- **Circular Resource Economy:** Complete water recycling system that captures overhead misting, filters, reheats/cools as necessary, and automatically pumps it back to the sprinkler heads.

3. Commercial Applications & Addressable Markets

Target Segment	Application / Opportunity	Why It Wins
Specialty Coffee Cultivation	Nurturing high-value Arabica coffee plants during vulnerable early-growth stages anywhere globally.	Protects growers against regional warming trends.
Agricultural Research Labs		

Target Segment	Application / Opportunity	Why It Wins
	Replacing expensive multi-million-dollar climate rooms with scalable, modular, mobile lab units.	Saves capital expenditure while isolating variables.
Urban & Vertical Farming	Deploying units in metropolitan warehouses or retail locations to display/grow flawless botanicals.	Ultra-compact footprint, mobility, and aesthetic appeal.
Biosecure Botanicals	Growing high-value pharmaceutical and cosmetic raw plant materials under sterile, pesticide-free environments.	Integrated continuous UV pathogen disinfection.

4. System Scalability & Business Model

We intend to monetize this patented technology through three primary pipelines:

1. **Direct Equipment Sales:** Standardized, scalable mobile nursery units targeted at commercial farms, agricultural labs, and urban operations.
2. **Service & Support Contracts:** Periodic replacement of UV sanitizers, water filters, and thermal calibration sensors.
3. **SaaS Integration:** Subscription-based access to proprietary environmental presets optimized for premium crop types.